

The Weltformel computation: the whole spectrum from $G, \hbar, c$ alone -- structure ab initio, absolute scale = Planck (the hierarchy is owed)

Ab-initio 3x4 fermion matrix (GeV)
scale = $M_{Pl} = 1.22\text{e}+19$ GeV (from $G, \hbar, c$)

up quark	u 4.29e+18	c 5.97e+18	t 1.22e+19
down quark	d 4.23e+18	s 5.89e+18	b 1.20e+19
charged lepton	e 7.63e+17	mu 1.06e+18	tau 2.17e+18
neutrino	nu1 5.83e+17	nu2 8.11e+17	nu3 1.66e+18
	gen I	gen II	gen III

Ab initio vs observed
structure right, scale = Planck (off by the hierarchy)

